SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 3 - Viva Based Questions

Course Code:	**DSA0405**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO4 –Poster Presentation**	**Assessment:**	**ASSESSMENT TOOL 3 - Viva Based Questions**
Weightage:	**40% of CIA**	**Total Marks:**	**100**
CO4 Statement:	Analyze industry-specific datasets using point estimation, confidence intervals, and hypothesis testing to infer population characteristics and make informed decisions. (BL4)		
Student Name:	Jeya Harshini R	**Reg. No.:**	192424051
Section / Batch:	2024	**Date:**	20.08.2026

Code of Conduct

I Jeya Harshini R (192424051) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Jeya Harshini R

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	**25**	**Thorough understanding; clearly identifies all key issues**	**Good understanding with most issues identified**	**Basic understanding; some issues missed**	**Poor understanding; problem not identified correctly**	
Application of Concepts	**25**	**Effectively applies relevant concepts to analyze the case deeply**	**Applies appropriate concepts with minor gaps**	**Limited application of concepts**	**Fails to apply relevant concepts**	

Analysis	**20**	**In-depth analysis with strong reasoning and insights**	**Good analysis with logical reasoning**	**Basic analysis with limited depth**	**Weak or no analysis; lacks reasoning**	
Solution Design	**20**	**Innovative, feasible solutions with strong justification**	**Logical solutions with adequate justification**	**Basic solutions with weak justification**	**Incorrect or no solution provided**	
Clarity	**10**	**Clear, well-structured, and easy to understand**	**Organized and understandable presentation**	**Limited clarity and structure**	**Poor clarity; difficult to understand**	

Here are **short viva-style answers** that are easy to remember and score with.

A. Introduction to Statistical Inference

1. What is statistical inference?

Statistical inference is the process of using sample data to draw conclusions about a population.

2. Why is statistical inference important in data science?

It helps make predictions, estimates, and decisions about large populations using sample data.

3. Difference between descriptive and inferential statistics?

- **Descriptive Statistics:** Summarizes collected data.
- **Inferential Statistics:** Draws conclusions about a population from a sample.

4. What is a population in statistics?

The complete set of individuals, objects, or observations under study.

5. What is a sample?

A subset of the population selected for analysis.

6. Why use samples instead of populations?

Because studying the entire population is costly, time-consuming, and sometimes impossible.

7. What is a population parameter?

A numerical characteristic of a population, such as μ, p, or σ^2.

8. What is a sample statistic?

A numerical value calculated from a sample, such as $\bar{x}$, $\hat{p}$, or s^2.

9. Real-world example of statistical inference?

A company surveys 500 customers and estimates the satisfaction level of all customers.

10. Two major activities in statistical inference?

1. Estimation
2. Hypothesis Testing

B. Frequentist Approach

11. What is the frequentist approach?

A statistical approach based on long-run frequencies of repeated samples.

12. Basic idea behind frequentist statistics?

Probability is defined as the long-run relative frequency of an event.

13. Difference between frequentist and Bayesian approach?

- Frequentist: Parameters are fixed.

- Bayesian: Parameters are treated as random variables.

14. What is fixed in frequentist statistics?

The population parameter is fixed.

15. What does long-run frequency mean?

The proportion of times an event occurs after many repeated trials.

16. Why is random sampling important?

It reduces bias and makes the sample representative of the population.

17. Can a population parameter be a random variable?

No. In the frequentist approach, population parameters are fixed but unknown.

18. Example of a frequentist problem?

Estimating the average monthly spending of all customers using a sample.

C. Point Estimation

19. What is a point estimate?

A single numerical value used to estimate a population parameter.

20. What is an estimator?

A rule or formula used to estimate a population parameter.

21. Difference between estimator and estimate?

- Estimator: Formula.
- Estimate: Numerical result obtained from the formula.

22. Point estimator for population mean?

Sample mean:

$$\bar{x}$$

23. How do you estimate a population proportion?

Using the sample proportion:

$$\hat{p} = \frac{x}{n}$$

24. Point estimate of population variance?

Sample variance:

$$s^2$$

25. Why is sample mean commonly used?

Because it is simple, unbiased, and efficient.

26. Properties of a good estimator?

- Unbiasedness
- Consistency
- Efficiency
- Sufficiency

27. What is an unbiased estimator?

An estimator whose expected value equals the true parameter.

28. What is bias of an estimator?

The difference between the estimator's expected value and the true parameter.

29. Difference between bias and variance?

- Bias: Accuracy.
- Variance: Consistency of estimates.

30. Why can different samples produce different estimates?

Because of sampling variability.

D. Variability in Estimates

31. Why does a point estimate have uncertainty?

Because it is based on a sample, not the entire population.

32. What is the sampling distribution?

The distribution of a statistic obtained from many samples.

33. What is standard error?

The standard deviation of a sampling distribution.

34. Difference between standard error and standard deviation?

- Standard deviation measures spread of data.
- Standard error measures spread of sample estimates.

35. Effect of increasing sample size on standard error?

Standard error decreases.

36. Why does variability decrease with larger samples?

Larger samples provide more information and more stable estimates.

37. Relationship between standard error and confidence interval?

Larger standard error leads to a wider confidence interval.

38. If two samples have different means, is one incorrect?

No. Different samples naturally produce different estimates.

E. Confidence Intervals

39. What is a confidence interval?

A range of values likely to contain the population parameter.

40. Why is a confidence interval better than a point estimate?

It provides both an estimate and its uncertainty.

41. What does a 95% confidence level mean?

In repeated sampling, 95% of constructed intervals will contain the true parameter.

42. Difference between confidence level and confidence interval?

- Confidence Level: Percentage (95%, 99%).
- Confidence Interval: Actual range of values.

43. Factors affecting confidence interval width?

- Sample size
- Standard deviation
- Confidence level

44. Effect of increasing sample size?

The confidence interval becomes narrower.

45. Effect of increasing confidence level from 95% to 99%?

The confidence interval becomes wider.

46. Difference between z and t confidence intervals?

- z-interval: Population SD known or large sample.
- t-interval: Population SD unknown and small sample.

47. When is the t-distribution used?

When population standard deviation is unknown and sample size is small.

F. Hypothesis Testing Using Confidence Intervals

48. What is hypothesis testing?

A method of making decisions about a population parameter using sample data.

49. What are H_0 and H_1?

- **H_0 (Null Hypothesis):** Assumes no effect or no difference.
- **H_1 (Alternative Hypothesis):** Assumes an effect or difference exists.

50. How can a confidence interval be used in hypothesis testing?

If the hypothesized value lies inside the confidence interval, fail to reject H_0. If it lies outside, reject H_0.
